

2 Kinematics

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Displacement is a vector change in position; velocity is its rate of change. A displacement-time gradient is velocity and velocity-time gradient is acceleration; signed area under velocity-time is displacement. The constant-acceleration equations apply only where acceleration remains uniform.

Worked example: follow the reasoning

A car moves from 4 to 16 m s⁻¹ in 6 s at constant acceleration: $a = (16 - 4)/6 = 2 \text{ m s}^{-2}$; $s = (4 + 16)6/2 = 60 \text{ m}$.

Quick check · multiple choice

A What is the gradient of a velocity-time graph? [1]

- A Displacement
- B Acceleration
- C Distance
- D Power

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A trolley accelerates uniformly from rest to 9 m s⁻¹ over 18 m. Find acceleration and travel time. [3]

Final answer: _____

Answers and explanation

Quick check: B. Acceleration

Question 1 [3 marks]

$v^2 = u^2 + 2as$, giving $a = 81/36 = 2.25 \text{ m s}^{-2}$ [2]; $t = (v - u)/a = 4.0 \text{ s}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.